Lecture 2

Quotients, products, and the symmetric group

Objectives. *Lecture 1 showed that the cosets G/H form a G-set but in general not a group. This lecture finds the subgroups for which they do, the normal ones, and with them quotient groups, the first isomorphism and correspondence theorems, and products. It then finds the conjugacy classes of S_n, indexed by partitions, and the sign.*

By the end of this lecture you will be able to:

- Decide whether a subgroup is normal, and identify a quotient by the first isomorphism theorem.
- For $N \trianglelefteq G$, list the subgroups between N and G from those of G/N.
- Recognise a group as a direct or semidirect product of a normal subgroup and a complement.
- Find the cycle type, class size, centraliser order and sign of a permutation, and write it as a product of transpositions.

2.1 Normal subgroups and quotients

In S_3 the rule $xH \cdot yH = xyH$ failed for $H = \langle (1\,2) \rangle$ (Example 1.13), because conjugation moves H: $(1\,3)(1\,2)(1\,3)^{-1} = (2\,3) \notin H$.

Definition 2.1 (Normal subgroup). A subgroup $N \leq G$ is *normal*, written $N \trianglelefteq G$, if $gNg^{-1} = N$ for every $g \in G$; that is, if $N_G(N) = G$ (Corollary 1.22).

It is enough that $gNg^{-1} \subseteq N$ for every g: applied to g^{-1} this gives $g^{-1}Ng \subseteq N$, that is, $N \subseteq gNg^{-1}$. Multiplying $gNg^{-1} = N$ on the right by g shows that normality is equivalent to $gN = Ng$ for every g. Every subgroup of an abelian group is normal, and so is $Z(G)$. The kernel of a homomorphism φ is normal: if $\varphi(k) = e$, then

$$\varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g)^{-1} = e.$$

Proposition 2.2 (Quotient groups). *Let $H \leq G$.*

(i) The rule $xH \cdot yH = xyH$ is well defined on G/H if and only if $H \trianglelefteq G$. Then G/H is a group, with identity H and $(xH)^{-1} = x^{-1}H$; and the map $q\colon G \to G/H$, $g \mapsto gH$, is a surjective homomorphism with kernel H.

(ii) $H \trianglelefteq G$ if and only if H is a union of conjugacy classes of G.

(iii) If $[G : H] = 2$, then $H \trianglelefteq G$.

Proof. Recall that $xH = yH$ exactly when $x^{-1}y \in H$.

(i) Let $H \trianglelefteq G$, and let $x' = xh$ and $y' = yk$ with $h, k \in H$. Then

$$x'y' = xy\,(y^{-1}hy)\,k$$

with $y^{-1}hy \in H$, so $x'y'H = xyH$.

Conversely, suppose the rule is well defined, and let $h \in H$ and $g \in G$. As $hH = eH$,

$$hg^{-1}H = (hH)(g^{-1}H) = (eH)(g^{-1}H) = g^{-1}H,$$

so $(g^{-1})^{-1}hg^{-1} = ghg^{-1} \in H$. Hence $gHg^{-1} \subseteq H$ for every g, which is enough for normality by the remark after the definition.

The group axioms in G/H follow from those in G, the rule says $q(xy) = q(x)q(y)$, and $q(g) = H$ exactly when $g \in H$.

(ii) $gHg^{-1} \subseteq H$ for all g says that with each h, H contains its class $[h]$.

(iii) The two left cosets partition G, so they are H and $G \setminus H$. Inversion gives a bijection from left cosets to right cosets, taking xH to Hx^{-1}. Thus the two right cosets are also H and $G \setminus H$. For each g, both gH and Hg contain g, so $gH = Hg$. Hence $gHg^{-1} = H$, and H is normal. ∎

Definition 2.3 (Quotient group and quotient map)**.** For $N \trianglelefteq G$, the group G/N of Proposition 2.2 is the *quotient group* of G by N. The homomorphism $q\colon G \to G/N$, $g \mapsto gN$, is the *quotient map*.

So the normal subgroups are exactly the kernels: N is the kernel of the quotient map $q\colon G \to G/N$.

Example 2.4 (A quotient of S_3)**.** Let $N = \langle (1\,2\,3) \rangle$. The two left cosets are

$$N = \{e, (1\,2\,3), (1\,3\,2)\}, \qquad (1\,2)N = \{(1\,2), (1\,3), (2\,3)\}.$$

The subgroup N has index 2, so it is normal. Its cosets therefore form a group. The identity is N, and the product of the other coset with itself is

$$((1\,2)N)^2 = (1\,2)^2 N = N.$$

These products determine S_3/N. It is isomorphic to C_2.

2.2 The isomorphism and correspondence theorems

Theorem 2.5 (Isomorphism theorems)**.** *Let $\varphi\colon G \to K$ be a homomorphism, $N \trianglelefteq G$, and $q\colon G \to G/N$ the quotient map.*

(i) (Factorisation.) If $N \leq \ker\varphi$, there is exactly one homomorphism $\bar{\varphi}\colon G/N \to K$ with $\bar{\varphi}(gN) = \varphi(g)$, that is, $\varphi = \bar{\varphi} \circ q$.

(ii) (First isomorphism theorem.) $G/\ker\varphi \cong \operatorname{im}\varphi$, by $g\ker\varphi \mapsto \varphi(g)$.

(iii) (Correspondence.) $H \mapsto q(H) = H/N$ is a bijection from the subgroups H of G that contain N to the subgroups of G/N, with inverse $\bar{H} \mapsto q^{-1}(\bar{H})$. It preserves inclusion and index, $[G : H] = [G/N : H/N]$, and $H \trianglelefteq G$ if and only if $H/N \trianglelefteq G/N$.

Proof. (i) If $gN = g'N$, write $g' = gn$ with $n \in N \leq \ker \varphi$. Then $\varphi(g') = \varphi(g)$, so $\bar{\varphi}$ is well defined. Multiplicativity follows from φ; surjectivity of q forces uniqueness.

(ii) Take $N = \ker \varphi$, normal because it is a kernel. The map $\bar{\varphi}$ from factorisation is onto $\operatorname{im} \varphi$. Its kernel is trivial, since $\bar{\varphi}(gN) = e$ exactly when $g \in N$, so it is injective (Lecture 1).

(iii) *The bijection.* Let $N \leq H \leq G$. If $q(g) = q(h)$ with $h \in H$, then $g \in hN \subseteq H$; so

$$q^{-1}(q(H)) = H.$$

For $\bar{H} \leq G/N$, $q^{-1}(\bar{H})$ is a subgroup containing $N = q^{-1}(\{N\})$, and $q(q^{-1}(\bar{H})) = \bar{H}$, as q is onto. So the two maps are inverse bijections, and both preserve inclusion.

The index. By the bijection,

$$gH = g'H \iff g^{-1}g' \in H \qquad \iff q(g)^{-1}q(g') \in q(H),$$

so $gH \mapsto q(g)\,q(H)$ is well defined and injective. It is also surjective, since q is. This gives a bijection $G/H \to (G/N)/(H/N)$. The latter is a set of cosets; it need not be a quotient group, since H need not be normal in G.

Normality. For $g \in G$, gHg^{-1} is a subgroup containing $gNg^{-1} = N$, and

$$q(gHg^{-1}) = q(g)\,q(H)\,q(g)^{-1}.$$

By the bijection, $gHg^{-1} = H$ exactly when $q(g)\,q(H)\,q(g)^{-1} = q(H)$; and $q(g)$ runs over all of G/N. ∎

Example 2.6 (The determinant of a triangle symmetry)**.** For the triangle group D_3 from Example 1.3, the determinant gives a homomorphism $\det\colon D_3 \to \{\pm 1\}$. The rotations have determinant 1 and the reflections have determinant -1. Thus

$$\ker \det = \langle r \rangle = \{e, r, r^2\}, \qquad \operatorname{im} \det = \{\pm 1\}.$$

The first isomorphism theorem identifies the quotient with the image:

$$D_3/\langle r \rangle \longrightarrow \{\pm 1\}, \qquad g\langle r \rangle \longmapsto \det g.$$

The determinant factors as the quotient map followed by this isomorphism. Thus a subgroup of $\{\pm 1\}$ and its corresponding subgroup of $D_3/\langle r \rangle$ have the same preimage in D_3, under the determinant and the quotient map respectively.

The two subgroups of $\{\pm 1\}$ are $\{1\}$ and $\{\pm 1\}$. Their preimages under the determinant are $\langle r \rangle$ and D_3. By correspondence, these are the only subgroups of D_3 containing $\langle r \rangle$. The reflection subgroup $\langle s \rangle$ does not contain $\langle r \rangle$, so it is not in this correspondence.

Lecture 6 lifts a representation by composing with q. Factorisation gives the converse when N lies in the kernel.

2.3 Direct and semidirect products

Definition 2.7 (Direct product)**.** The *direct product* of groups G and H is the group $G \times H$ of pairs (g, h), with multiplication $(g, h)(g', h') = (gg', hh')$.

The identity is (e, e) and the inverse of (g, h) is (g^{-1}, h^{-1}). Associativity follows from that in the two factors.

For finite G, H, $(g,h)^k = e$ exactly when both $o(g)$ and $o(h)$ divide k. Thus $o((g,h)) = \operatorname{lcm}(o(g), o(h))$. In $C_m \times C_n$, a pair of generators has order $\operatorname{lcm}(m,n)$, and every element has order dividing it. A group of order mn is cyclic exactly when it has an element of order mn. Thus the product is cyclic exactly when $\operatorname{lcm}(m,n) = mn$, or equivalently $\gcd(m,n) = 1$.

Example 2.8 (Two products of cyclic groups)**.** The pair of generators in $C_2 \times C_3$ has order 6, so

$$C_2 \times C_3 \cong C_6,$$

whereas every element of $C_2 \times C_2$ has order at most 2, so that product is not cyclic.

We now give conditions under which a group is a product of two of its subgroups.

Theorem 2.9 (Semidirect products)**.** *Let $N \trianglelefteq G$ and $H \leq G$ with $N \cap H = \mathbf{1}$.*

(i) $NH = \{nh : n \in N,\ h \in H\}$ is a subgroup. The map $(n,h) \mapsto nh$ is a bijection $N \times H \to NH$. If N and H are finite, $|NH| = |N|\,|H|$.

(ii) In NH, with $hn'h^{-1} \in N$,

$$(nh)(n'h') = (n \cdot hn'h^{-1})(hh').$$

(iii) If moreover $H \trianglelefteq G$, then $nh = hn$ for all $n \in N$, $h \in H$, and $(n,h) \mapsto nh$ is an isomorphism $N \times H \to NH$.

If G is finite and $|N|\,|H| = |G|$, then $NH = G$.

Proof. (i) Insert $h^{-1}h$:

$$nh\,n'h' = n\,(hn'h^{-1})\,hh'.$$

Since $hn'h^{-1} \in N$, this proves closure under products. Also $e \in NH$, and

$$(nh)^{-1} = h^{-1}n^{-1} = (h^{-1}n^{-1}h)\,h^{-1} \in NH.$$

If $nh = n'h'$, then $n'^{-1}n = h'h^{-1}$ lies in $N \cap H = \mathbf{1}$, so $n = n'$ and $h = h'$.

(ii) The same insertion gives the stated multiplication rule.

(iii) Group the element $nhn^{-1}h^{-1}$ in two ways:

$$nhn^{-1}h^{-1} = n\,(hn^{-1}h^{-1}) \in N,$$
$$nhn^{-1}h^{-1} = (nhn^{-1})\,h^{-1} \in H.$$

So $nhn^{-1}h^{-1} = e$, and $nh = hn$. The multiplication formula becomes $(nh)(n'h') = (nn')(hh')$. Thus $(n,h) \mapsto nh$ is a homomorphism, and it is bijective by unique factorisation.

Finally, if G is finite and $|N|\,|H| = |G|$, the subgroup NH is contained in G and has the same order, so $NH = G$. ∎

Definition 2.10 (Internal semidirect product)**.** Let $N \trianglelefteq G$ and $H \leq G$, with $N \cap H = \mathbf{1}$ and $NH = G$. Then G is the *internal semidirect product* $N \rtimes H$, and H is a *complement* to N. If H is normal too, this is an *internal direct product*.

If H is normal too, Theorem 2.9 gives $G \cong N \times H$.

The multiplication rule uses only N, H and conjugation, $n' \mapsto hn'h^{-1}$. Lecture 3 uses the counting form, $|N|\,|H| = |G|$, for groups of order pq.

Example 2.11 (The Klein four-group)**.** The *Klein four-group* V_4 consists of the following permutations in S_4:

$$V_4 = \{e,\ (1\,2)(3\,4),\ (1\,3)(2\,4),\ (1\,4)(2\,3)\}.$$

Write the three nonidentity permutations as a, b and c, in the order shown. Each has order 2, and

$$ab = (1\,2)(3\,4)\,(1\,3)(2\,4) = (1\,4)(2\,3) = c.$$

Since $c^{-1} = c$, taking inverses gives $ba = c$ as well. Then $ac = b$ and $bc = a$, in either order. Thus V_4 is closed under products and inverses, and is an abelian subgroup of S_4. Its subgroups $\langle(1\,2)(3\,4)\rangle$ and $\langle(1\,3)(2\,4)\rangle$ are normal in V_4. They meet in $\mathbf{1}$, and their orders have product $2 \cdot 2 = 4$. The direct-product criterion gives $V_4 \cong C_2 \times C_2$.

Example 2.12 (The dihedral group as a semidirect product)**.** Let $n \geq 3$. By Example 1.18, the rotation subgroup $\langle r\rangle$ has index 2 in D_n, so it is normal by Proposition 2.2. It meets the reflection subgroup $\langle s\rangle$ only in e: rotations have determinant 1 and s has determinant -1. Their orders multiply to $2n = |D_n|$, so

$$D_n = \langle r\rangle \rtimes \langle s\rangle, \qquad \langle r\rangle \cong C_n, \quad \langle s\rangle \cong C_2.$$

Here the semidirect product uses the action of s by inversion. Indeed, $srs^{-1} = r^{-1}$ by Example 1.3, so for $a, b \in \mathbb{Z}$ and $\epsilon, \delta \in \{0, 1\}$ the multiplication rule is

$$(r^a s^\epsilon)(r^b s^\delta) = r^{a+(-1)^\epsilon b} s^{\epsilon+\delta}.$$

The exponents are read modulo n and 2, respectively. For $n \geq 3$, $r \neq r^{-1}$, so $rs \neq sr$. Hence D_n is not isomorphic to the abelian group $C_n \times C_2$.

2.4 S_n: cycles, conjugacy and z_μ

The entry sets of the disjoint cycles of $\sigma \in S_n$ (Lecture 1) are the orbits of $\langle\sigma\rangle$ on $\{1, \ldots, n\}$, with a one-cycle for each fixed point.

> **Proposition 2.13** (Cycle decomposition)**.** *Let $\sigma \in S_n$ and $a \in \{1, \ldots, n\}$, and let $k \geq 1$ be least with $\sigma^k(a) = a$. The orbit of a under $\langle\sigma\rangle$ is $\{a, \sigma(a), \ldots, \sigma^{k-1}(a)\}$, with k distinct elements. So σ is the product of the cycles $(a\ \sigma(a)\ \ldots\ \sigma^{k-1}(a))$, one for each orbit, and these commute.*

Proof. Such a k exists, as $\sigma^{o(\sigma)} = e$. If $\sigma^i(a) = \sigma^j(a)$ with $0 \leq i < j < k$, then $\sigma^{j-i}(a) = a$, against the choice of k; and $\sigma^m(a)$ depends only on m modulo k. So the orbit is as stated, and σ acts on it as the cycle.

Every point lies in one orbit, and only its own cycle moves it, so σ and the product of the cycles agree at every point. Cycles that move disjoint sets commute, for the same reason. ∎

The decomposition is unique up to the order of the cycles and the starting point of each. Indeed, the support of each cycle in any disjoint-cycle decomposition is an orbit, and the orbits are determined by σ.

Example 2.14 (Finding the cycles)**.** Let $\sigma \in S_5$ send

$$1 \mapsto 3, \quad 2 \mapsto 5, \quad 3 \mapsto 1, \quad 4 \mapsto 4, \quad 5 \mapsto 2.$$

Starting at 1 gives $1 \mapsto 3 \mapsto 1$. The first unused point is 2, and $2 \mapsto 5 \mapsto 2$. The remaining point 4 is fixed. Thus

$$\sigma = (1\,3)(2\,5)(4) = (1\,3)(2\,5).$$

The last equality omits the one-cycle, as in Lecture 1.

Definition 2.15 (Partitions and cycle type)**.** A *partition* $\lambda \vdash n$ is a sequence $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_\ell > 0$ of integers with sum n. Its *multiplicity* $m_i(\lambda)$ is the number of parts equal to i; we write $(2, 1^2) = (2, 1, 1)$; and $p(n)$ is the number of partitions of n.
The *cycle type* of $\sigma \in S_n$ is the partition μ of its cycle lengths, fixed points counted as cycles of length 1. For a partition μ,

$$z_\mu = \prod_{i \geq 1} i^{m_i(\mu)}\, m_i(\mu)!\,.$$

For example, the classes of S_3 found in Example 1.21 have cycle types (1^3), $(2, 1)$ and (3). For type $(2, 1)$ the multiplicities are $m_1 = m_2 = 1$, so $z_{(2,1)} = 2$. The class-size formula below gives $3!/2 = 3$ elements, as in that example.

A cycle of length i has order i, and the cycles of σ commute and move disjoint sets, so $\sigma^k = e$ exactly when every part of μ divides k:

$$o(\sigma) = \operatorname{lcm}(\mu_1, \ldots, \mu_\ell).$$

Theorem 2.16 (Classes of S_n)**.**

(i) *For $\tau \in S_n$,*

$$\tau\,(a_1\, a_2\, \ldots\, a_k)\,\tau^{-1} = (\tau(a_1)\,\tau(a_2)\,\ldots\,\tau(a_k)).$$

(ii) *$\sigma, \sigma' \in S_n$ are conjugate if and only if they have the same cycle type. So the classes of S_n correspond to the partitions of n, and there are $p(n)$ of them.*

(iii) *The class of type μ has $n!/z_\mu$ elements, and $\left|C_{S_n}(\sigma)\right| = z_\mu$ for σ of type μ.*

Proof. (i) Let $\sigma = (a_1\,\ldots\,a_k)$, indices read modulo k. Then

$$\tau\sigma\tau^{-1}\colon\ \tau(a_i) \longmapsto \tau(a_{i+1}),$$

and $\tau\sigma\tau^{-1}$ fixes $\tau(b)$ for every other b, since σ fixes b.

(ii) Conjugation relabels entries and keeps lengths, by the cycle formula. Conversely, if σ and σ' have the same type, write their cycles, fixed points included, σ' under σ with equal lengths aligned:

$$\sigma = (a_1\,\ldots\,a_k)(b_1\,\ldots\,b_l)\cdots,$$
$$\sigma' = (a'_1\,\ldots\,a'_k)(b'_1\,\ldots\,b'_l)\cdots.$$

The map τ sending each entry to the one below it is a permutation, and $\tau\sigma\tau^{-1} = \sigma'$ by that formula. Every partition occurs as a cycle type, by the filling construction below.

(iii) Fill the pattern of μ, one cycle of each part's length, with $1, \ldots, n$ in each of the $n!$ ways. Each filling gives a permutation of type μ, and each permutation of type μ arises.

The cycle slots of the pattern are ordered. By uniqueness of the decomposition, two fillings give the same σ exactly when they differ by rotations of cycles and exchanges of

cycles of equal length. A cycle of length i has i starting points, and the $m_i(\mu)$ slots for cycles of length i can be exchanged in $m_i(\mu)!$ ways. This includes the slots for fixed points. Thus each permutation comes from z_μ distinct fillings. So, with Theorem 1.20,

$$|[\sigma]| = \frac{n!}{z_\mu},$$
$$\left|C_{S_n}(\sigma)\right| = \frac{n!}{|[\sigma]|} = z_\mu. \quad \blacksquare$$

For example, type $(2, 1^2)$ in S_4 has

$$z_{(2,1^2)} = 2 \cdot 2! = 4, \qquad \frac{4!}{z_{(2,1^2)}} = 6.$$

For the permutation $(1\,2)$, the four fillings of the ordered pattern $(_\ _)(_)(_)$ are

$$(1\,2)(3)(4), \qquad (2\,1)(3)(4),$$
$$(1\,2)(4)(3), \qquad (2\,1)(4)(3).$$

The factor 2 counts the starting points of the two-cycle; the factor 2! counts the ways to exchange the fixed-point slots.

The inverse of a cycle is the reversed cycle,

$$(a_1\, a_2\, \ldots\, a_k)^{-1} = (a_k\, \ldots\, a_2\, a_1),$$

so σ^{-1} has the type of σ: every permutation is conjugate to its inverse. Lecture 10 finds a conjugating element that fixes n.

2.5 Sign, generators and A_n

Definition 2.17 (Permutation matrix)**.** The *permutation matrix* of $\sigma \in S_n$ is the $n \times n$ matrix $D(\sigma)$ with entries $D(\sigma)_{ij} = \delta_{i,\sigma(j)}$.

Let $e_1, \ldots, e_n$ be the standard basis of $\mathbb{C}^n$; the subscript distinguishes a basis vector e_j from the identity element e. The jth column of $D(\sigma)$ is $e_{\sigma(j)}$. For $\tau \in S_n$ and $1 \le j \le n$,

$$D(\sigma)D(\tau)e_j = D(\sigma)e_{\tau(j)} = e_{\sigma\tau(j)} = D(\sigma\tau)e_j,$$

so $D(\sigma\tau) = D(\sigma)D(\tau)$.

Definition 2.18 (Sign)**.** The *sign* of $\sigma \in S_n$ is $\operatorname{sgn}\sigma = \det D(\sigma)$. A permutation is *even* if its sign is $+1$ and *odd* if its sign is -1.

For example,

$$D((1\,2)) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \operatorname{sgn}(1\,2) = -1 \quad \text{in } S_3.$$

Recall from Lecture 1 that a transposition is a 2-cycle.

Theorem 2.19 (Sign).

(i) $\operatorname{sgn}\colon S_n \to \{\pm 1\}$ *is a homomorphism, and the sign of a transposition is* -1.

(ii) *A* k*-cycle is a product of* $k-1$ *transpositions,*

$$(a_1\, a_2\, \dots\, a_k) = (a_1\, a_k)(a_1\, a_{k-1}) \cdots (a_1\, a_2),$$

with the empty product understood as e *when* $k = 1$. *Thus* $\operatorname{sgn}\sigma = (-1)^{n-c(\sigma)}$, *where* $c(\sigma)$ *is the number of cycles of* σ, *fixed points included.*

(iii) *For* $n \geq 2$, *the* alternating group $A_n = \ker \operatorname{sgn}$ *is a normal subgroup of index* 2, *with* $n!/2$ *elements.*

Proof. (i) The determinant is multiplicative, so sgn is a homomorphism to $\mathbb{C}^\times$. The columns of $D(\sigma)$ are distinct standard basis vectors, so $D(\sigma)^{\mathsf{T}} D(\sigma) = \mathbb{1}$. Taking determinants gives $(\det D(\sigma))^2 = 1$, so $\operatorname{sgn}\sigma = \pm 1$.

$D((i\, j))$ is the identity matrix with columns i and j exchanged, and exchanging two columns changes the sign of a determinant.

(ii) For $k = 1$ the factorisation is the empty product. Let $k \geq 2$. Read the product right to left. Each a_i is fixed by every factor that does not contain it, so

$$a_1 \mapsto a_2, \qquad a_i \mapsto a_1 \mapsto a_{i+1} \quad (2 \leq i < k), \qquad a_k \mapsto a_1.$$

By multiplicativity a k-cycle has sign $(-1)^{k-1}$. For type μ,

$$\operatorname{sgn}\sigma = (-1)^{\sum_j (\mu_j - 1)} = (-1)^{n - c(\sigma)}.$$

(iii) For $n \geq 2$, $\operatorname{sgn}(1\,2) = -1$, so sgn is onto $\{\pm 1\}$, and $S_n/A_n \cong \{\pm 1\}$ by Theorem 2.5. ∎

The sign is constant on classes: $\operatorname{sgn}(\tau\sigma\tau^{-1}) = \operatorname{sgn}\sigma$ because $\{\pm 1\}$ is abelian. For $n = 1$ it is trivial, so the index-two assertion needs $n \geq 2$. In S_3 the even permutations are e, $(1\,2\,3)$ and $(1\,3\,2)$, so $A_3 = \langle (1\,2\,3) \rangle$. Lecture 3 uses index-two normality and the union-of-classes criterion to rule out a subgroup of order 6 in A_4.

We write s_i for the adjacent transpositions. The subscript distinguishes s_i from the reflection s in the dihedral example.

Proposition 2.20 (Generators). *S_n is generated by its transpositions, and also by $s_1, \dots, s_{n-1}$, where $s_i = (i\ i{+}1)$.*

Proof. The cycle decomposition and the factorisation of a cycle into transpositions in Theorem 2.19 prove the first claim. For the second it is enough to reach each $(i\, j)$ with $i < j$. If $j = i + 1$, this is s_i. Otherwise let

$$w = s_{j-1} s_{j-2} \cdots s_{i+1}.$$

Right to left, its factors carry $i+1$ to $i+2, \dots, j$, so $w(i+1) = j$, while $w(i) = i$. Conjugation relabels the entries of a cycle (Theorem 2.16), so

$$w s_i w^{-1} = (w(i)\ w(i{+}1)) = (i\, j).$$

For $n = 1$ the empty set generates $S_1 = \{e\}$. ∎

The s_i are the *Coxeter generators*. Lecture 11 and Lecture 12 describe a representation of S_n by the matrices of $s_1, \dots, s_{n-1}$ alone, which determine it by Proposition 2.20.

2.6 Worked example

Find the classes, normal subgroups and product structure of S_4.

Cycle types (Theorem 2.16) and signs (Theorem 2.19) give:

type μ	representative	z_μ	class size $24/z_\mu$	order	sign
(1^4)	e	24	1	1	+1
$(2,1^2)$	$(1\,2)$	4	6	2	−1
(2^2)	$(1\,2)(3\,4)$	8	3	2	+1
$(3,1)$	$(1\,2\,3)$	3	8	3	+1
(4)	$(1\,2\,3\,4)$	4	6	4	−1

Example 2.21 (Guard: class size is not element order)**.** The 3 elements of type (2^2) and the 6 of type $(2,1^2)$ all have order 2: the order is the least common multiple of the parts.

Normal subgroups. A normal subgroup is a union of classes containing e (Proposition 2.2), of order dividing 24. The other class sizes are $3, 6, 6, 8$. Subtracting 1 from the divisors of 24 and checking which are sums of these sizes gives $0, 3, 11, 23$. So the possible orders are

$$\{1, 4, 12, 24\},$$

with a unique union of classes for each order: 3 and 11 occur only as 3 and $3+8$. All four are realised by subgroups. The group V_4 is e with the class (2^2) (Example 2.11); A_4 consists of the even types. So the normal subgroups of S_4 are $\mathbf{1}$, V_4, A_4, S_4.

The quotient. S_4 permutes the three ways of splitting $\{1,2,3,4\}$ into two pairs,

$$P_1 = \{12\,|\,34\}, \qquad P_2 = \{13\,|\,24\}, \qquad P_3 = \{14\,|\,23\}.$$

Numbering P_k as k, this is a homomorphism $\pi\colon S_4 \to S_3$ (Proposition 1.10). For instance,

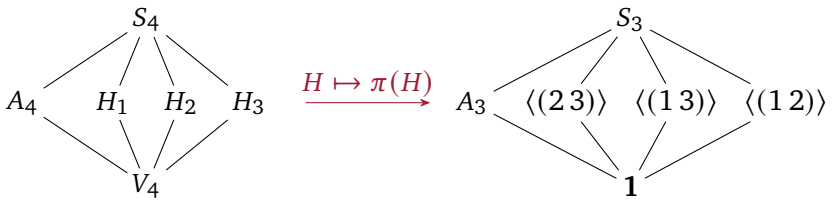
$$\pi((1\,2)) = (2\,3), \qquad \pi((1\,2\,3)) = (1\,3\,2).$$

Each element of V_4 fixes every P_k, so $\ker\pi$ is a normal subgroup containing V_4 but not $(1\,2\,3)$. By the list of normal subgroups above, the kernel is V_4. So π has $24/4 = 6$ values, is onto, and $S_4/V_4 \cong S_3$ (Theorem 2.5).

The correspondence. By the correspondence theorem (Theorem 2.5), the subgroups of S_4 containing V_4 correspond to those of S_3: $\mathbf{1}$, the three $\langle(i\,j)\rangle$, A_3 and S_3 (Corollary 1.17). The preimage of $\langle(2\,3)\rangle$, the stabiliser of 1, is the stabiliser H_1 of P_1, of order 8:

$$H_1 = V_4 \cup \{(1\,2),\ (3\,4),\ (1\,3\,2\,4),\ (1\,4\,2\,3)\}.$$

The preimage of the stabiliser of k is the stabiliser H_k of P_k for each $k = 1, 2, 3$. By correspondence, the preimage of A_3 has index 2 in S_4. It is normal, and the list above identifies it as A_4. Every subgroup of index 2 is normal, so A_4 is the only one. In the diagram, lines indicate inclusions and corresponding subgroups occupy matching positions:



Lecture 3 meets H_1, H_2, H_3 again as the subgroups of order 8, the largest power of 2 dividing 24.

The product. Let $S_3 \leq S_4$ be the stabiliser of 4. Each $v \neq e$ in V_4 moves 4, so $V_4 \cap S_3 = \mathbf{1}$; and $|V_4|\,|S_3| = 24$. So $S_4 = V_4 \rtimes S_3$ (Theorem 2.9). These factors do not form an internal direct product. The stabiliser S_3 contains only three of the six transpositions, which form one conjugacy class in S_4, so it is not normal.

Example 2.22 (Guard: normality is not transitive)**.** The subgroup $\langle (1\,2)(3\,4) \rangle$ is normal in V_4, as V_4 is abelian. And $V_4 \trianglelefteq A_4$, since $V_4 \trianglelefteq S_4$ and $V_4 \leq A_4$. But the relabelling formula (Theorem 2.16) gives

$$(1\,2\,3)\,(1\,2)(3\,4)\,(1\,2\,3)^{-1} = (2\,3)(1\,4),$$

so $\langle (1\,2)(3\,4) \rangle$ is not normal in A_4. In the chain of subgroups used in Lecture 4, $S_4 \trianglerighteq A_4 \trianglerighteq V_4 \trianglerighteq \langle (1\,2)(3\,4) \rangle \trianglerighteq \mathbf{1}$, each term need only be normal in the one before.

Self-check. Find the size of the class of $(1\,2)(3\,4)$ in S_5 using its cycle type. Explain the factor 2! in the answer.

Answer: The type is $(2^2, 1)$, so $z_{(2^2,1)} = 2^2 \cdot 2! = 8$ and the class has $5!/8 = 15$ elements. The factor 2! counts the exchanges of the two cycles of length 2 in a filling.

Summary.

- *The main results:* G/N is a group exactly when N is normal (Proposition 2.2); $G/\ker\varphi \cong \operatorname{im}\varphi$, and the subgroups of G/N correspond to those of G above N (Theorem 2.5); $N \cap H = \mathbf{1}$ with N normal gives $NH = N \rtimes H$ (Theorem 2.9); the classes of S_n are the cycle types, with centralisers of order z_μ (Theorem 2.16).
- *Handed on:* quotients and the isomorphism theorems; the sign and A_n; the generators s_i; $S_4/V_4 \cong S_3$ and $S_4 = V_4 \rtimes S_3$.
- *Next:* Lecture 3 asks what the order of a group forces about its subgroups: Sylow's theorems.

References

Dummit and Foote, *Abstract Algebra*: normal subgroups and quotients, §3.1, Prop. 5, p. 81, Thm 6 and Prop. 7, p. 82; index 2, §3.2, p. 91; the first and fourth (correspondence) isomorphism theorems, Thm 16, p. 97, and Thm 20, p. 99; sign and A_n, §3.5, pp. 107–109; classes of S_n, §4.3, Props. 10–11, pp. 125–126; direct and semidirect products, §5.4, Thm 9, p. 171, §5.5, Thm 12, p. 180. Rotman, *An Introduction to the Theory of Groups*: cycles and sign, Thms 1.1–1.6, pp. 6–9; the correspondence theorem, Thm 2.28, p. 38; conjugacy in S_n, Thm 3.5, p. 47; semidirect products, pp. 167–170. Sagan, *The Symmetric Group*: z_λ and the class size, Prop. 1.1.1 and (1.2), p. 3.